

Experiment No. 4:

To determine the temperature coefficient of resistance of the material of a wire.

4.1 Objectives

To find the temperature coefficient of resistance of a given wire using Meter Bridge.

4.2 Prelab

Student must have clear idea about the objectives and related theory of the experiment. Lab Demonstrator will confirm that the students have submitted their previous group lab reports in printed form before giving them the attendance. The marks of performance depend greatly on this part.

4.3 Outcomes:

After completing this lab work student will be able to answer the following questions:

- What do you mean by temperature coefficient of resistance?
- What is Meter bridge? What is the principle of working of a Meter Bridge?
- How resistance changes with temperature? Is there any positive or negative temperature coefficient of resistance?

4.4 Timing and Lab Activities (Total 3 Hours)

- Lab Preparation (15 minutes):
 - For attendance and equipment distribution among the groups.
- Lecture on Theory (30 minutes):
 - Teacher will clarify the objective and theory of the experiment.
- Lecture on Procedure (15 minutes):
 - Demonstrator will make the students understand the procedure of the experiment.
- Experimental Work (90 to 100 minutes):
 - Design of experiment: 10 minutes
 - Data collection: 30 minutes
 - Data analysis, drawing graphs and completing the result part: 50 to 60 minutes
- Post Lab Discussion (15 to 20 minutes):
 - Teacher will summarize the total lab work and have a discussion with the students related with the questions given in the outcomes part.
- Report Submission:
 - Student will prepare their lab reports as groups in MS word within one week.
 - Students must submit their group lab reports as printed before the starting of the next lab.

4.5 Theory

Temperature coefficient of resistance is the property of the material of a substance. It is the measure of change in electrical resistance of any substance per degree of temperature change.

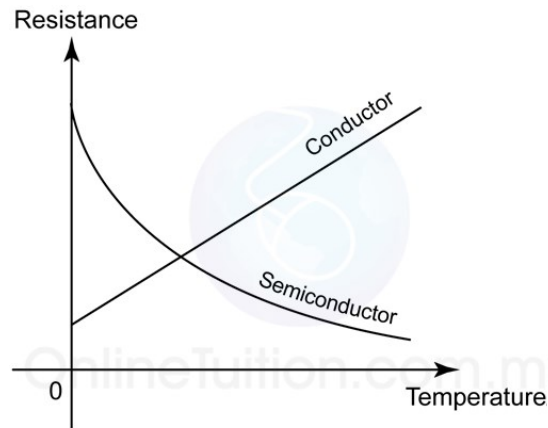


Figure 4.1: A graph that shows the changes in resistances of a conductor and a semiconductor with temperature. For conductor the resistance increases linearly and for semiconductor the resistance decreases exponentially with the temperature.

For a conductor the resistance increases with increase of temperature, as the figure 5.1 shows. If R_1 is the resistance of a conductor at temperature θ_1 and at a higher temperature θ_2 the resistance raises to R_2 , then we can write

$$R_2 = R_1[1 + \alpha(\theta_2 - \theta_1)], \quad (1)$$

where α is the temperature coefficient of the material of the conductor.

Rearranging Eq. 1, we get

$$\begin{aligned} R_2 &= R_1 + \alpha R_1(\theta_2 - \theta_1) \\ \Rightarrow \alpha R_1(\theta_2 - \theta_1) &= R_2 - R_1 \\ \Rightarrow \alpha &= \frac{R_2 - R_1}{R_1(\theta_2 - \theta_1)} = \frac{\Delta R}{R_1 \Delta \theta}, \quad (2) \end{aligned}$$

where $\Delta R (= R_2 - R_1)$ is the change in resistance due to the change in temperature $\Delta \theta (= \theta_2 - \theta_1)$.

In Eq. (2) if we put $R_1 = 1 \text{ Ohm}$ and $\Delta \theta = 1^\circ \text{C}$ we get $\alpha = \Delta R$, thus we define the temperature coefficient of resistance of a substance as the change in resistance per unit resistance (per ohm) for unit change in temperature (per $^\circ \text{C}$). Also we find the unit of α as per $^\circ \text{C}$.

For a conductor, if we know the resistances R_1 at a temperature θ_1 and R_2 at a higher temperature θ_2 , then we can calculate the temperature coefficient of resistance of its material by using the Eq. 2. The unknown resistance can be determined by a meter bridge that works with the Wheatstone bridge principle.

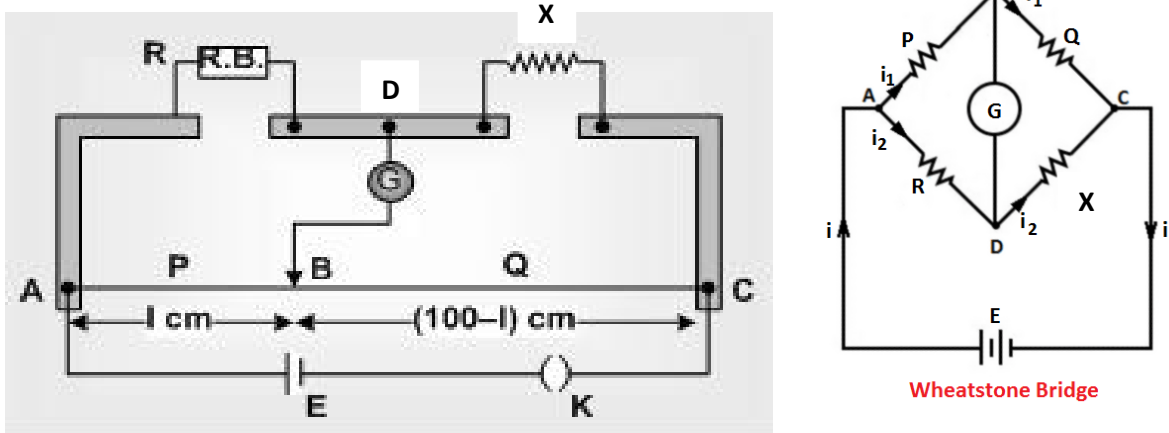


Figure 4.2: A meter bridge can be used to determine an unknown resistance by using the Wheatstone bridge principle.

In the meter bridge circuit as in fig. 5.2, for the null deflection in the galvanometer we get

$$\frac{P}{Q} = \frac{R}{X}$$

$$\Rightarrow X = \frac{Q}{P} \times R = \frac{\rho(100 - l)/A}{\rho l/A} \times R = \frac{100 - l}{l} \times R, \quad (3)$$

where ρ is the specific resistance of the material and A is the cross sectional area of the wire of the meter bridge.

According to Eq. (3) if we know the length of the balance point, l and the resistance, R we can determine the unknown resistance X .

4.6 Equipment

Power supply, meter bridge, galvanometer, jockey, resistance box, coil of conducting wire, commutator, thermometer, Beaker, water, electric heater, etc.

4.7 Circuit Diagram for the experiment

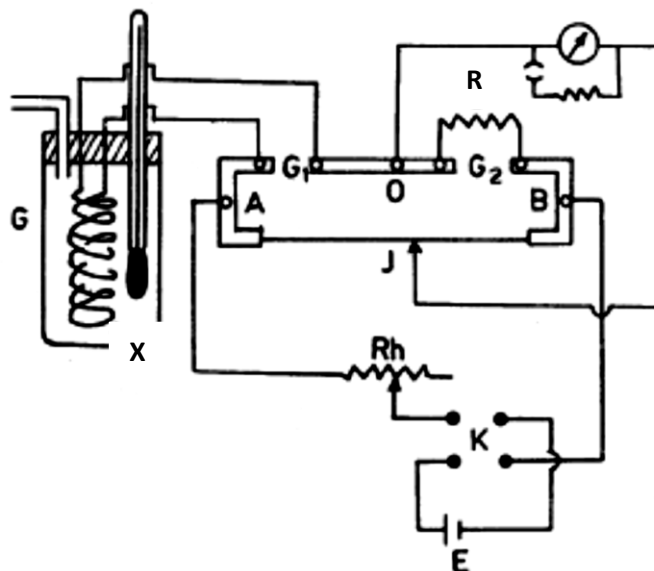


Figure 4.3: Circuit diagram for the experiment.

4.8 Procedure

- Set up the apparatus as the circuit diagram shows in figure 4.3.
- First connect the coil of the wire in the left gap G_1 and keep it inside the Beaker at room temperature. Note the temperature, θ_1 from the thermometer. Connect the resistance box in the right gap G_2 of the meter bridge. By closing the nob's in different positions of the commutator you can change the direction of current flow in the circuit. Take different resistances (say 5, 7, 9 ohm) from the resistance box to find the position of the balance points on the wire of the meter bridge for direct current as well as reverse current.
- Now Connect the coil of the wire in the right gap G_2 and the resistance box in the left gap G_1 . Find the balance points for direct current and reverse current for 5, 7 and 9 ohms.
- Pour hot water in the Beaker to raise the temperature. Note the high temperature, θ_2 from the thermometer. For different resistances (5, 7 and 9 ohms) find the balance points for direct current as well as reverse current connecting the coil of the wire in the left gap then in the right gap.
- Calculate the value of unknown resistances for left gap and right gap by using the two equations given in the calculation part. Also calculate the temperature coefficient, α .

4.9 Experimental Data

Table 4.1: Readings for the balance points on meter bridge.

Temperature (°C)	Known resistance R (Ohms)	Position of unknown resistance X	Position of Balance point, l (cm)		Mean l (cm)	100 – l (cm)	X (Ohms)	Mean X (Ohms)
			Direct current	Reverse current				
$\theta_1 =$		Left						$X_1 =$
		Right						
		Left						
		Right						
		Left						
		Right						
$\theta_2 =$		Left						$X_2 =$
		Right						
		Left						
		Right						
		Left						
		Right						

4.10 Calculation

- For left gap: $X = \frac{l}{100-l} \times R$
- For right gap: $X = \frac{100-l}{l} \times R$
- Temperature coefficient of resistance, $\alpha = \frac{X_2 - X_1}{X_1(\theta_2 - \theta_1)}$

4.11 Result

Temperature ($^{\circ}\text{C}$)	Resistance (ohm)	Temperature coefficient of resistance ($/^{\circ}\text{C}$)

4.12 Resources

For further understanding students may go through the following resources:

- Fundamental of Physics (10th Edition): Resistance and resistivity (Chapter 26, page 755)
- Video Links:
 - Temperature coefficient: <https://www.youtube.com/watch?v=TgmOfi2rn0s>
 - Meter Bridge: <https://www.youtube.com/watch?v=nqx8vIdHVkQ>